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CONTINUATION OF NOTES

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1 PACKAGES, MACROS AND GRAPHICS

1.1 PACKAGES

There are different packages that are required in order for the compiler to run perfectly. They are usually included in the preamble that is between `\begin{document}` and `\documentclass`

- The backslash geometry is important as it helps customize every page as without it there will be a very large space left between the page label and the end of the page. you can however customize it to your own settings as done earlier.
- full page to change the format of the document ranging from width length to point of beginning of the document
- `amsmath` - Advanced mathematical equations; provides `align`, `cases`, `gather`, `split`, etc.-used to align two or more equations
- `amssymb` - Extra mathematical symbols such as $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$... special symbols like $\therefore$, $\because$, $\#$
- `amsfonts` - Additional mathematical fonts; mainly used for commands like `\mathbb{b}`.
- `amsthm` - Creates theorem, lemma, corollary, proposition, and definition environments.
- `mathtools` - Extends `amsmath` with additional symbols and equation-formatting tools. `graphicx` - Inserts and resizes images.
- `float` - Controls the placement of figures and tables using options like `[H]`.
- `array` - Improves table and matrix formatting. especially where you need to place your figures center, left or right.
- `booktabs` - Produces professional-looking tables using `toprule`, `midrule`, and `bottomrule`. these are much cleaner than repeated `\hline`

- multirow - Allows cells to span multiple rows in a table.
- geometry - Adjusts page margins and page dimensions.
- fancyhdr - Creates custom headers and footers. mostly used for assignments and reports
- hyperref - Makes references, citations, URLs, and the table of contents clickable.
- cleveref - Automatically formats references (Equation, Figure, Theorem, etc.).
- biblatex - Handles bibliographies and citations. manages references too...referring to einstein
- xcolor - Adds colours to text, tables, and other elements.
- listings - Displays source code with formatting.
- minted - Displays source code with advanced syntax highlighting.it is like writting a code
- tikz - Creates diagrams, graphs, flowcharts, vectors, and illustrations. eg tikzpicture or tikzcircuit
it is extremely powerful
- pgfplots - Draws mathematical graphs and plots.
- siunitx - Formats scientific units and measurements correctly. eg
- mhchem - Typesets chemical equations and formulas. eg
- multicol - Creates multiple columns of text.
- textcomp - Provides extra text symbols such as ° and €for additional symbols like the euro
- algorithm - Creates algorithm environments.
- algpseudocode - Writes pseudocode inside algorithm environments.
- enumitem - Customizes bullet and numbered lists.
- setspace - Controls line spacing (single, 1.5, double).
- caption - Customizes figure and table captions.
- subcaption - Creates subfigures and subtables.
- titlesec - Customizes section and chapter titles.

- appendix - Helps organize appendices.
- csquotes - Provides intelligent quotation marks and quotation environments.

NOTES

This is the symbol for the set of all real numbers; $\mathbb{R}$

This is the symbol for the set integers; $\mathbb{Z}$

This is the symbol of for the set if rational numbers; $\mathbb{Q}$

1.2 MACROS

A $\text{\LaTeX}$ macros is a user defined command that replaces a piece of text or code with a shorter command, allowing repetitive content to be written more efficiently and consistently throughout a document. It saves you time, make your document easy to maintain and avoid repeatedly typing the same code. it acts as a shortcut or abbreviation

example; $x \in \mathbb{R}$

for an equation; $y = \frac{x}{3x^2+x+1}$ we referred our equation as we defined it earlier at the preamble

1.3 GRAPHICS

in this case you use this command to insert images to you pdf. usually the package graphicx is used then for the command, type backslash include graphics for scale choose any betwee 0 and 1. then insert your image name say limit. you may or may not include the png or jnx format of the image file.

2 ATTEMPTED WRITING EXERCISE

1. Is it possible for a sequence to converge to two different numbers? if so, give an example why.
2. Explain how to use partial sums to determine if a series converges or diverges. Give an example why.
3. explain why $\int_1^{\infty} f(x) dx$ and $\sum_{n=1}^{\infty} a_n$ need to converge to the same value, even if they are both convergent.
4. in your own words explain the meaning of the alternating series Remainder Theorem. How is this theorem be useful?
5. Explain the difference between absolute and conditional convergence. Give an example of each.
6. The ratio Test is inconclusive if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$. Give an example of one convergent series and one divergent series for which $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$. Explain your example (in example six you can use the backslash opening square brackets and backslash closing square brackets instead of the backslash display style)

3 ERRORS AND DEBUGGING

Identify your errors from the red marked section each time it pops up. This includes function errors, brackets errors or even math mode errors. The compiler usually helps you to identify from the line number.

4 COMMENTS

Just like c there are comments in $\text{\LaTeX}$ and there are outlined so that the computer does not have to compile them.

However in $\text{\LaTeX}$ they are identified using a percentage symbol. Examrrior hence i have left it out the line has bee left out since has an error

make sure that there are no blank lipile;